

Toward Principled Threshold Selection for Blink Region Detection in Biological Signals

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- **ABSTRACT** This study evaluated an epoch-aware pipeline for EEG blink-region detection that avoided estimating detection thresholds from concatenated blink-containing and blink-free epochs. The three-stage pipeline first screened non-overlapping epochs by peak-to-peak amplitude using an autoreject-inspired cross-validation criterion, then estimated a sample-level threshold from the suspicious-epoch pool using either a mean-based or median/MAD-based estimator with a fallback for small pools, and finally extracted blink regions and mapped them to epoch-local coordinates. BLINKER-concat, MNE-annot, Proposed-Mean, and Proposed-Med were compared on the Internal and Cao2018 naturalistic driving-EEG corpora using event-level overlap matching and macro-averaged F_1 . At 30 s, Proposed-Med achieved a pooled macro- F_1 of 84.03%, significantly exceeding BLINKER-concat at 72.56% and MNE-annot at 60.86%. Proposed-Mean and Proposed-Med were statistically indistinguishable ($p = 1$), which indicated that the gain came from epoch screening rather than the choice of Stage-B estimator. The channel-selection ablation further showed that a single frontopolar electrode was sufficient for the strongest single-channel detection on both corpora. Finally, pooled macro- F_1 varied by only 0.56 percentage points across epoch durations from 10 s to 120 s, and no tested duration differed from the 30 s reference after Bonferroni correction.
- **INDEX TERMS** blink detection, electroencephalography, epoch-aware detection, naturalistic driving, threshold selection

I. INTRODUCTION

Blinking is a physiological process that involves the closure and reopening of the eyelids [1]. Its primary functions are to protect the ocular surface, maintain corneal moisture, and support visual comfort [1]. Beyond these physiological functions, blink behaviour has been increasingly recognised as an important indicator in fatigue-driving research [2, 3]. Variations in blink-related features, such as blink frequency, blink duration, eye-closure duration, and blink patterns, can reflect changes in alertness and cognitive state [2, 4]. As fatigue increases, drivers may exhibit slower or prolonged blinks, which can indicate reduced vigilance and a higher risk of impaired driving performance [2]. Therefore, blink behaviour provides a useful and non-invasive marker for monitoring driver fatigue [3, 5].

In the neuroscience study of fatigue driving, blink activity can be captured through external methods, such as camera-based eye monitoring, or through electrophysiological recordings, particularly electroencephalography and electrooculography [3, 5]. Although EEG is mainly used to measure brain activity, blink-related components are often present in EEG signals because eyelid movements and eye movements generate strong electrical potentials [6, 7]. These ocular activities are especially prominent in frontal EEG electrodes

due to their close proximity to the eyes [6, 8]. As a result, blink information can be extracted from EEG recordings, either by analysing the ocular components embedded in the EEG signal or by combining EEG with EOG-based measurements [5, 9]. This makes EEG/EOG recordings valuable not only for studying neural activity during fatigue but also for identifying blink-related markers associated with drowsiness [2, 10].

It is important to distinguish between two major perspectives on blink activity in the existing literature [3, 6]. In many EEG studies, blink-related activity is treated as an artifact because it contaminates neural signals and may interfere with the interpretation of brain activity [6, 7]. In such cases, blink components are commonly removed during EEG preprocessing [6, 11]. However, in fatigue-driving studies, blink activity can also be treated as meaningful information rather than noise [3, 5]. From this perspective, blink events are identified, extracted, and analysed as potential indicators of driver fatigue [5, 10]. Blink detection methods can generally be classified into machine-learning-based, morphology-based, and thresholding-based approaches [9, 12]. Machine learning methods rely on trained models to classify blink events, while morphology-based methods identify blinks based on their characteristic waveform shape [13, 14]. Thresholding-based methods, in contrast, detect candidate blink events when the

signal amplitude exceeds a certain threshold value, as blink activity typically produces large-amplitude deflections, especially in frontal channels [8, 15].

Thresholding is one of the most straightforward and widely used approaches for detecting blink activity from EEG or EOG signals [15, 16]. The simplicity of thresholding makes it computationally efficient and easy to implement, particularly in real-time fatigue-detection systems [16, 17]. Existing approaches range from single- or few-channel detectors that apply amplitude thresholds with rule-based decision logic to hybrid pipelines that combine thresholding as an initial detection step with additional candidate-refinement mechanisms [8, 9, 12, 14–16, 18–24]. Despite these developments, selecting an appropriate threshold remains challenging because blink amplitude varies across participants, electrode locations, recording devices, signal quality, and experimental conditions [20, 24]. Blink amplitude can vary across participants, electrode locations, recording devices, signal quality, and experimental conditions [20, 24]. A threshold that is too low may produce false detections due to noise or other artifacts, whereas a threshold that is too high may miss genuine blink events [16, 17]. This backdrop explains why simple one-value global thresholds can be attractive, yet brittle. Therefore, adaptive and robust threshold selection remains a critical consideration for reliable blink detection, particularly in fatigue-driving applications [5, 10].

In many fatigue-driving studies, EEG recordings are segmented into epochs that correspond to specific task events or analysis windows [25]. This epoch-structured setting introduces additional challenges for threshold-based blink detection [16, 17]. When thresholds are estimated from the entire recording, they may be biased by the presence of clean epochs that contain few or no blinks, which may distort threshold estimates and suppress blink detection in epochs where blinks are more prevalent [11, 24]. As a result, naive application of thresholding in an epoch-structured context may risk under-detection of blinks, which undermines the reliability of fatigue assessments based on blink metrics [5, 10]. Addressing this challenge requires developing blink detection methods that explicitly account for the epoch structure and can adapt threshold estimation to the specific characteristics of each epoch or subset of epochs, rather than relying on a global threshold derived from the entire recording [9, 24].

This paper makes several contributions to blink detection in fatigue driving research. First, we propose a double thresholding blink detection pipeline comprising epoch screening, sample level threshold estimation, and blink region extraction. To our knowledge, this is among the first frameworks to integrate these three stages explicitly to address the challenges associated with epoch structured EEG recordings. Second, we systematically compare the proposed pipeline against baseline methods, including BLINKER concat [8] and MNE annot. To support reproducibility and facilitate further research, we also developed an automated pipeline for extracting ocular indices from EEG recordings and made it freely available as a Python toolbox called *pyblinker* (TODO).

The toolbox is intended to facilitate reproducible blink analysis and encourage more consistent reporting of ocular indices in EEG based fatigue research.

The remainder of the paper is organized as follows. Section Methods and materials describes the steps and methods used to extract blinks from EEG data, as well as the datasets we used to validate the results. Section Results presents the results of running the BLINKER toolbox on a large and varied collection of EEG data. Section The BLINKER software briefly introduces the toolbox, referring readers to the more detailed online documentation and downloads. Section Visual verification of BLINKER output describes our manual video verification process and some interesting results we obtained when comparing subject behavior captured in video with blink signals captured by EEG. Section Discussion offers some concluding remarks and discusses limitations of the method.

II. METHOD AND MATERIALS

a: Processing pipeline overview.

A. Processing Pipeline Overview

The proposed method is an epoch-aware, double-thresholding pipeline that transforms a continuous multichannel EEG recording into channel-wise blink-region detections through five sequential operations. First, the EEG signals are downsampled and band-pass filtered before being divided into fixed-length, non-overlapping epochs. The proposed double-thresholding method then operates through three detection stages. Stage A performs multichannel peak-to-peak screening to identify candidate epochs containing prominent amplitude excursions. Stage B pools the samples from these candidate epochs to estimate a robust sample-level threshold independently for each channel. Stage C applies the resulting channel-specific thresholds to all valid epochs to localize blink regions and determine their epoch identifier, channel, onset time, and duration. Figure 1 summarizes the complete processing pipeline, while Stages A, B, and C are described in detail in the following subsections.

b: Preprocessing

First, the continuous recording is downsampled to a common sampling grid of 100 Hz. Standardising the sampling rate makes the amplitude thresholds and the minimum-duration gate comparable across recordings acquired on different hardware, and reduces the per-sample computational cost of the downstream stages without discarding any information needed to resolve a blink, whose spectral content lies well below the new Nyquist frequency.

Second, a zero-phase band-pass filter from 1.0 Hz to 20.0 Hz is applied which removes slow drifts and DC offsets at the low end and high-frequency muscle and line noise at the high end, isolating the low-frequency, high-amplitude deflection that characterises an eye blink.

Third, each filtered recording is split into non-overlapping, fixed-length epochs. Epoching is what makes the thresholds

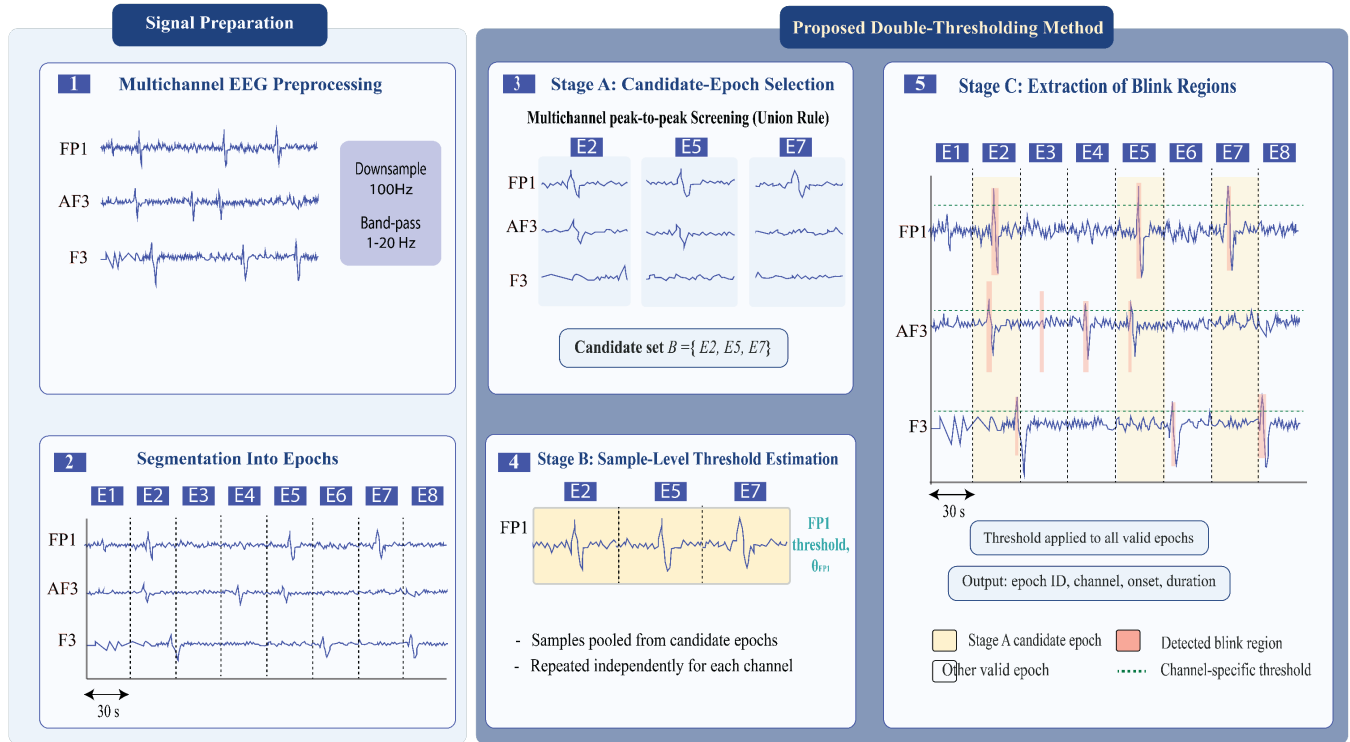


FIGURE 1. Overview of the proposed epoch-aware double-thresholding pipeline for EEG blink-region detection. Continuous multichannel EEG signals are downsampled to 100 Hz, band-pass filtered from 1 to 20 Hz, and divided into non-overlapping 30-s epochs. Stage A applies multichannel peak-to-peak screening and a union rule to select candidate epochs, represented by E2, E5, and E7 in this example. Stage B pools the samples from the selected epochs to estimate a robust sample-level threshold independently for each channel. Stage C applies the resulting channel-specific thresholds to all valid epochs and reports each detected blink region by its epoch identifier, channel, onset time, and duration. Pale yellow denotes Stage A candidate epochs, red denotes detected blink regions, and the dashed green lines represent the channel-specific thresholds.

adaptive, rather than deriving a single level for the entire session, the pipeline treats each short epoch as a local statistical context.

c: Selection of Candidate Epochs

The first stage, Stage A, of the proposed pipeline performs an initial epoch-level screening using an *autoreject*-inspired threshold-selection procedure to identify epoch that are most likely to contain prominent blink-related activity. Rather than relying on a single channel, the screening decision integrates channel-level information across multiple EOG/ EEG channels, allowing an epoch to be selected when abnormal amplitude excursions are detected in one or more channels. At this stage, the objective is not to localize individual blink samples, but to identify candidate epochs containing prominent amplitude activity. This multi-channel screening strategy therefore narrows the search space to potentially blink-related epochs for more detailed sample-level analysis in the subsequent detection stages.

Let $X \in \mathbb{R}^{N \times C \times T}$ denote the epoched multichannel signal after preprocessing, where N is the number of valid epochs, C is the number of channels, and T is the number of samples per epoch. The sample at epoch $i \in \mathcal{E} = \{1, \dots, N\}$, channel $c \in \mathcal{C} = \{1, \dots, C\}$, and within-epoch time index $t \in \{1, \dots, T\}$ is denoted by $x_{i,c,t}$.

For each epoch and channel, an amplitude-range statistic is computed using the peak-to-peak value

$$a_{i,c} = \max_{1 \leq t \leq T} x_{i,c,t} - \min_{1 \leq t \leq T} x_{i,c,t}.$$

Large values of $a_{i,c}$ indicate that the corresponding epoch contains a high-amplitude excursion in channel c , a pattern consistent with ocular or other transient artifacts. Because signal amplitudes may vary across channels, participants, and recording conditions, a fixed peak-to-peak threshold is not imposed. Instead, a channel-specific threshold τ_c^* is calculated from each of the channel's. Candidate thresholds are taken from the empirical range of peak-to-peak amplitudes for that channel, denoted $\mathcal{Q}_c = \{a_{i,c} : i \in \mathcal{E}\}$, and are evaluated by a cross-validation criterion that measures the discrepancy between retained training epochs and validation epochs.

After the channel-specific thresholds, τ_c^* , have been estimated, the peak-to-peak amplitude of each epoch is evaluated separately for each EEG channel against its corresponding threshold. An epoch is retained as a candidate when the amplitude in at least one channel exceeds the threshold τ_c^* defined for that channel. Thus, the epoch-level screening decision incorporates information across multiple channels using a union rule, rather than relying on a single predefined channel. The resulting set of candidate epochs is defined as

The set of suspicious epochs is therefore

$$\mathcal{B} = \{i \in \mathcal{E} : b_i = 1\}.$$

where b_i denotes the epoch-level screening outcome indicating that epoch (i) satisfies the selection criterion. These candidate epochs represent high-amplitude segments that may contain prominent blink-related activity and are retained for subsequent sample-level blink-region detection.

d: Computation of Sample-Level Blink-Region Threshold Candidates

Stage B estimates channel-specific sample-level amplitude thresholds using signal samples drawn from the candidate epochs $\mathcal{B}$ identified in Stage A. The estimation combines a measure of central tendency with a robust measure of dispersion and a sensitivity parameter, allowing the threshold to adapt to the amplitude characteristics of each channel while limiting the influence of isolated extreme samples.

Let $\mathcal{S}$ denote the set of epochs used for estimating the sample-level threshold. In the primary analysis, $\mathcal{S} = \mathcal{B}$, so that the estimate is driven by epochs already identified as likely to contain blink-related activity.

For each channel c , all samples from the selected epochs are pooled into a one-dimensional empirical sample set

$$\mathcal{Y}_c = \{x_{i,c,t} : i \in \mathcal{S}, 1 \leq t \leq T\}.$$

The sample-level threshold is estimated using a robust location-plus-dispersion form. Let ξ_c be the channel-specific center of the sample distribution. The default choice is the median,

$$\xi_c = \text{median}_{y \in \mathcal{Y}_c} y,$$

It should be note that, the arithmetic mean may be substituted when a mean-centered threshold is required for comparison.

The dispersion is estimated by the median absolute deviation (MAD),

$$\text{MAD}_c = \text{median}_{y \in \mathcal{Y}_c} |y - \text{median}_{u \in \mathcal{Y}_c} u|,$$

and is rescaled to a standard-deviation-compatible quantity by

$$\sigma_c^{\text{rob}} = 1.4826 \text{ MAD}_c.$$

The constant 1.4826 is the usual normal-consistency factor for the MAD.

For a sensitivity parameter $k > 0$, the resulting sample-level blink-region threshold candidate for channel c is

$$\theta_c = \xi_c + k\sigma_c^{\text{rob}}.$$

The set

$$\Theta = \{\theta_c : c \in \mathcal{C}\}$$

constitutes the channel-wise threshold candidates passed to the blink-region extraction stage.

A. EXTRACTION OF BLINK REGIONS

The final stage applies the sample-level thresholds $\Theta = \{\theta_c\}_{c=1}^C$ obtained in Stage B to obtain channel-wise blink regions. Specifically, blink regions are searched for within each valid epoch $i \in \mathcal{E} = \{1, \dots, N\}$ and each channel $c \in \mathcal{C}$, by comparing the samples $x_{i,c,t}$ of that epoch and channel against the threshold candidate θ_c of the same channel.

The final output of this stage is therefore a channel-indexed collection of blink regions, each represented by an epoch identifier, a channel label, an onset time, and duration.

B. EXPERIMENTAL SETUP

The evaluation protocol was designed to compare all detector conditions on the same recordings, reference annotations, and scoring criteria. The datasets, event-level performance measures, and statistical analyses are described below.

1) Datasets

Two EEG corpora acquired during sustained simulated-driving tasks were used, providing 104 complete recording sessions in total. The corpora differed in their acquisition montages and data provenance, allowing the proposed method to be evaluated under different hardware setups.

The public Cao2018 corpus contains 62 sessions from 27 participants who performed an approximately 90-minute lane-keeping task in an immersive driving simulator [25]. The present study included the 58 sessions marked as complete and excluded the remaining four incomplete sessions. This study used the raw recordings. All 32 recorded electrophysiological channels (30 scalp electrodes and the A1/A2 mastoid reference channels) were supplied to the pipeline; the concurrently recorded vehicle-position channel was excluded. These 32 channels were Fp1, F3, Fp2, F4, FT7, FC3, C3, FC4, FT8, C4, TP7, CP3, TP8, CP4, T5, P3, O1, T6, P4, O2, F7, T3, Fz, FCz, Cz, Pz, Oz, F8, T4, CPz, A1, and A2.

The Internal corpus comprised 46 complete sessions recorded during a simulated lane-keeping task. EEG was acquired with a 128-channel EGI HydroCel system, with CPz as the reference. Although the original recordings used a high-density montage, the present analyses retained a predefined subset of 32 electrodes corresponding to conventional 10–20 scalp locations, following the subsampled 128-to-32-channel montage of [26]: Fp1, AF3, F3, Fp2, AF4, F4, FC5, FC1, C3, FC2, FC6, C4, CP5, CP1, CP2, CP6, P5, P3, PO3, O1, PO4, P4, P7, O2, F7, T7, Fz, Cz, Pz, Oz, F8, and T8.

We divided the 32 electrodes into four anatomical regions (frontal, central, parietal, and occipital) and further divided each region into left and right hemispheres, as per [27], as shown in Figure 2. In the figure, the resulting eight regions (frontal left, frontal right, central left, central right, parietal left, parietal right, occipital left, and occipital right) are abbreviated FL, FR, CL, CR, PL, PR, OL, and OR, respectively. The figure shows the scalp locations of the 32 electrodes: the blue circles represent electrodes located in both the Internal and Cao2018 montages, the yellow circles represent elec-

trodes located only in the Internal montage, and the red circles represent electrodes located only in the Cao2018 montage.

Of these eight regions, only the frontal, central, parietal, and occipital regions (left and right) were used in the analysis. The following electrodes were not assigned to any lateralized region and were likewise excluded from the analysis: F7, T7, F8, T8, Fz, FCz, Cz, CPz, Pz, and Oz.

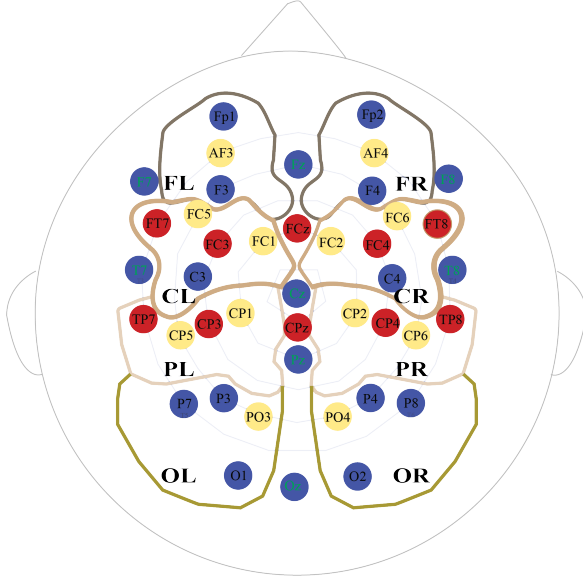


FIGURE 2. Scalp locations of the 32 electrodes used in this study, divided into eight anatomical regions (frontal left/right, central left/right, parietal left/right, and occipital left/right, abbreviated FL, FR, CL, CR, PL, PR, OL, and OR). Blue circles mark electrodes present in both the Internal and Cao2018 montages, yellow circles mark electrodes present only in the Internal montage, and red circles mark electrodes present only in the Cao2018 montage.

2) Evaluation Metrics

a: Event matching.

Performance was evaluated at the blink-event level rather than at the sample level. For a predicted interval P and a reference interval G , temporal overlap was quantified using intersection over union (IoU):

$$\text{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|}. \quad (1)$$

Here, $|\cdot|$ denotes interval duration. A predicted interval was eligible for a match when its IoU with an unmatched reference interval was at least 0.1. Matching was one-to-one, so a predicted or reference interval could contribute to at most one match. Matched predictions were counted as true positives (TP), remaining predictions as false positives (FP), and remaining reference intervals as false negatives (FN).

b: Session-level measures.

Precision, recall, and F_1 were calculated independently for each session s from the event counts:

$$\text{Precision}_s = \frac{\text{TP}_s}{\text{TP}_s + \text{FP}_s}, \quad (2)$$

$$\text{Recall}_s = \frac{\text{TP}_s}{\text{TP}_s + \text{FN}_s}, \quad (3)$$

$$F_{1,s} = \frac{2 \text{ Precision}_s \text{ Recall}_s}{\text{Precision}_s + \text{Recall}_s}. \quad (4)$$

Event-level F_1 was the primary outcome because blink events were sparse; sample-level accuracy would be dominated by non-blink samples and would therefore provide limited information about event-detection performance.

c: Macro-averaging.

For a measure $M \in \{\text{Precision}, \text{Recall}, F_1\}$, the reported macro-average over S sessions was

$$\bar{M} = \frac{1}{S} \sum_{s=1}^S M_s. \quad (5)$$

Thus, each session contributed equally, irrespective of its number of annotated blinks. Pooled results across the two corpora used the same unweighted average over all 104 sessions; event counts were not pooled before computing the measures. This approach prevents sessions with many blinks from dominating the headline result, as would occur with micro-averaging.

d: Operating-point selection.

Unless an analysis explicitly considered a fixed single electrode, each tested channel configuration was summarised at a retrospective *best-channel-per-session* operating point. Within each session, the output electrode with the highest event-level F_1 was selected, and its associated precision, recall, and F_1 were retained for macro-averaging. The same rule was applied to every detector condition in a comparison. Because this selection used the reference annotations, it represents an oracle operating point and should not be interpreted as prospective, label-free channel selection.

3) Statistical Analysis

a: Paired comparisons.

The recording session was the unit of analysis. Differences between detector conditions were assessed using the Wilcoxon signed-rank test applied to matched pairs of session-level F_1 scores. This non-parametric paired test was selected because F_1 is bounded and no normality assumption was imposed on the paired differences. All inferential tests reported in the main analyses were two-sided, and zero-valued paired differences were omitted according to the Wilcoxon convention.

b: Multiplicity and effect size.

The family-wise significance level was set to $\alpha = 0.05$. Bonferroni adjustment was applied separately within each pre-specified family of related comparisons. The four-condition strategy analysis comprised six pairwise tests, whereas the epoch-duration analysis comprised six comparisons between the non-reference durations and the 30 s reference. Channel-subset comparisons were corrected separately within each corpus over the 13 subsets tested against the full 32-electrode montage. Analysis-specific family sizes for any additional comparisons are stated with the corresponding results. Bonferroni-adjusted p -values are reported together with the matched-pairs rank-biserial correlation r as an effect-size measure; the sign of r indicates the direction of the paired difference in the stated comparison.

III. RESULTS

A. EXPERIMENT 1: EEG CHANNEL-SUBSET ANALYSIS

In the proposed pipeline, the Stage A screened epochs using a union rule over the channels provided to the pipeline, so changing the channel set, in theory, may also change which epochs reached Stage B. Restricting the available channels into pipeline, therefore changed the detector itself rather than merely changing the signals presented to a fixed detector. Therefore, the channel-subset analysis was performed by re-running the complete pipeline with each channel subset rather than evaluating a detector trained on a different channel set.

1) Full-Montage Reference

First, we supply the complete 32 channel montage to the pipeline, which serves as a reference for the subsequent channel-subset analysis. This also interesting as we assume no prior knowledge of where brain region projected blinks the most, and the full montage is the most general configuration.

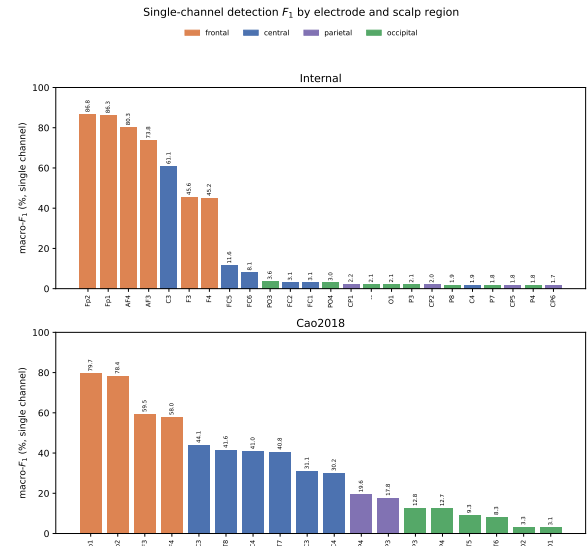
Against this full-montage reference, per-electrode detection quality was concentrated at the frontopolar pair rather than distributed across the frontal region, with each electrode scored within the 32-channel Proposed-Med run in which Stage A screened epochs using every electrode (Figure 3). On Internal, Fp2 and Fp1 reached F_1 values of 86.82% and 86.32%, respectively, before performance fell at the neighbouring frontal sites AF4 and AF3 to 80.32% and 73.76%, with F3, and F4 at 45.65%, and 45.23%; the lowest value was 1.75% at CP6.

On Cao2018, Fp1 and Fp2 reached F_1 values of 79.68% and 78.45%, respectively, whereas the neighbouring frontal sites F3 and F4 were lower at 59.51% and 57.99%; the lowest value was 3.10% at O1.

The central electrodes also showed a dataset-specific asymmetry. On Internal, C3 reached $F_1 = 61.06\%$ while C4 reached $F_1 = 1.85\%$, whereas on Cao2018 C3 and C4 were close at $F_1 = 31.08\%$ and $F_1 = 30.18\%$, respectively, so the pronounced C3–C4 asymmetry was specific to Internal.

Consistent with the preceding per-electrode pattern, the region-collapsed results showed a strong anterior-to-posterior decline in Proposed-Med performance, although the decline

was not strictly monotonic on Internal because occipital F_1 slightly exceeded parietal F_1 (Table 2 and Figure 3). On Internal, frontal electrodes ($n = 4$) yielded 75.64% precision, 56.76% recall, and 61.24% F_1 , compared with 36.26%, 12.88%, and 14.80%, respectively, for central electrodes ($n = 6$), 14.09%, 1.10%, and 1.94% for parietal electrodes ($n = 4$), and 12.47%, 1.38%, and 2.30% for occipital electrodes ($n = 8$). On Cao2018, the decrease was monotonic from frontal ($n = 2$; 72.17% precision, 54.05% recall, 58.75% F_1) to central ($n = 6$; 62.57%, 31.23%, 38.13%), parietal ($n = 2$; 48.70%, 13.32%, 18.70%), and occipital electrodes ($n = 6$; 34.04%, 5.29%, 8.24%), with both precision and recall decreasing across successive regions. When frontal and frontopolar electrodes were combined, their mean F_1 was 69.68% across $n = 6$ electrodes on Internal, compared with 8.05% across the $n = 26$ other electrodes, while on Cao2018 the corresponding means were 68.91% across $n = 4$ frontal and frontopolar electrodes and 23.95% across the $n = 28$ other electrodes.



Performance was aggregated by selecting the best channel for each session and then averaging over sessions. Comparisons with the full montage used two-tailed Wilcoxon signed-rank tests with Bonferroni correction within each dataset, and matched-pairs rank-biserial correlation was reported as the effect size. Because Stages A–C were rerun using each subset alone, each subset constituted a self-contained detector, so this design measured whether a subset was sufficient on its own; it was not a leave-one-region-out analysis and did not establish that any region was necessary to the complete system. Internal electrodes were reported by their 10–20 scalp locations.

At the anatomical-subset level, Proposed-Med retained performance comparable to the full montage only for frontal configurations, whereas central and more posterior subsets performed substantially worse (Table 1; Figure 4). The Frontal subset was statistically indistinguishable from the full montage on both corpora, with $F_1 = 86.76\%$, a paired difference of -1.49 percentage points, and $p = 1.000$ on Internal, and $F_1 = 79.61\%$, a paired difference of -1.06 percentage points, and $p = 1.000$ on Cao2018. After Bonferroni correction, Frontal, Frontal-Left and Frontal-Right did not differ significantly from the full montage on Internal, with $p = 1.000$, $p = 0.070$ and $p = 0.113$, respectively. On Cao2018, only Frontal and Frontal-Left remained non-significant, with $p = 1.000$ and $p = 0.123$, whereas Frontal-Right differed significantly from the full montage, with $p = 0.001$.

As expected, performance fell steeply when only central, parietal, occipital or posterior subsets were used. Internal yielded $F_1 = 58.59\%$ for Central, $F_1 = 2.64\%$ for Parietal, $F_1 = 4.06\%$ for Occipital and $F_1 = 4.26\%$ for Posterior, while Cao2018 yielded $F_1 = 46.68\%$, $F_1 = 16.68\%$, $F_1 = 11.97\%$ and $F_1 = 17.36\%$ for the corresponding subsets. All differences for these central, parietal, occipital and posterior subsets remained significant after correction, with large matched-pairs rank-biserial effect sizes ($p < 0.001$ in every case; for Central, $r = -1.00$ on Internal and $r = -0.99$ on Cao2018).

In contrast to the frontal subsets, the non-frontal subsets showed markedly lower F_1 than the full montage in both Internal and Cao2018 (Table 1). On Cao2018, the parietal, occipital, and posterior subsets reached F_1 values of 16.68% , 11.97% , and 17.36% , respectively, whereas the same subsets reached only 2.64% , 4.06% , and 4.26% on Internal.

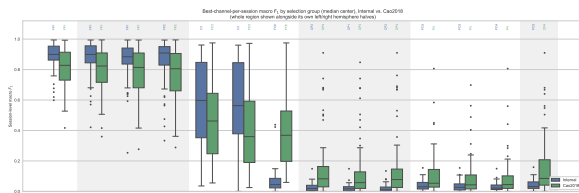


FIGURE 4. Session-level macro F_1 by channel-selection group (median centre), Internal versus Cao2018.

TABLE 1. Experiment 1 channel-subset performance of Proposed-Med (median centre) against the full-montage reference. Each subset is a self-contained detector: Stage A, Stage B and Stage C were all re-run on that channel subset alone. F_1 is macro-averaged over sessions at the best-channel-per-session operating point and reported as a percentage; ΔF_1 is the paired difference from the full montage in percentage points, tested with a two-tailed Wilcoxon signed-rank test and Bonferroni-corrected over the 13 subsets compared within each dataset, with the matched-pairs rank-biserial correlation r as effect size. n is the number of electrodes in the subset; the frontal subset contains AF3 and AF4 on Internal but not on Cao2018, whose montage lacks them.

Subset	n	F_1 (%)	ΔF_1 (pp)	p	r
<i>Internal (EGI 128, 46 sessions)</i>					
All (full montage)	32	88.25	+0.00	–	–
Frontal	6	86.76	−1.49	1.000	−0.07
Frontal-L	3	85.18	−3.07	0.070	−0.49
Frontal-R	3	84.59	−3.66	0.113	−0.45
Central	6	58.59	−29.66	< 0.001	−1.00
Central-L	3	57.67	−30.58	< 0.001	−1.00
Central-R	3	7.14	−81.11	< 0.001	−1.00
Parietal	4	2.64	−85.61	< 0.001	−1.00
Parietal-L	2	2.17	−86.09	< 0.001	−1.00
Parietal-R	2	2.12	−86.13	< 0.001	−1.00
Occipital	8	4.06	−84.19	< 0.001	−1.00
Occipital-L	4	3.43	−84.82	< 0.001	−1.00
Occipital-R	4	3.04	−85.21	< 0.001	−1.00
Posterior	12	4.26	−83.99	< 0.001	−1.00
<i>Cao2018 (10–20, 58 sessions)</i>					
All (full montage)	32	80.68	+0.00	–	–
Frontal	4	79.61	−1.06	1.000	−0.20
Frontal-L	2	78.41	−2.27	0.123	−0.40
Frontal-R	2	76.77	−3.90	0.001	−0.59
Central	6	46.68	−33.99	< 0.001	−0.99
Central-L	3	41.42	−39.26	< 0.001	−1.00
Central-R	3	41.01	−39.67	< 0.001	−1.00
Parietal	2	16.68	−64.00	< 0.001	−1.00
Parietal-L	1	14.80	−65.88	< 0.001	−1.00
Parietal-R	1	16.22	−64.46	< 0.001	−1.00
Occipital	6	11.97	−68.71	< 0.001	−1.00
Occipital-L	3	9.92	−70.75	< 0.001	−1.00
Occipital-R	3	9.68	−70.99	< 0.001	−1.00
Posterior	8	17.36	−63.31	< 0.001	−1.00

3) Single-Channel Operation

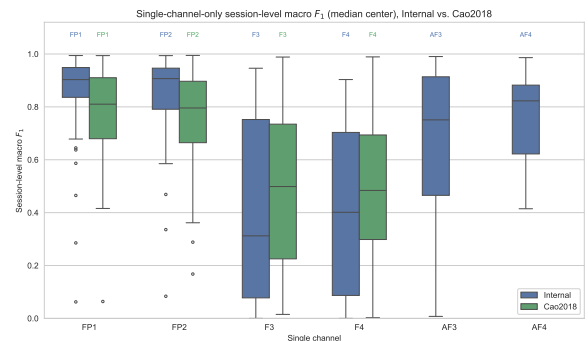


FIGURE 5. Session-level macro F_1 for the single-electrode selection groups, in which the complete pipeline was run on one electrode alone.

Following the channel-subset results above, Experiment 1 further tested whether a single electrode was sufficient as a self-contained detector by running the complete Stage A–Stage C Proposed-Med pipeline separately on each single-

electrode selection in both Internal and Cao2018, using 30-second epochs and the median centre. Internal comprised 46 sessions and 6 single-electrode selections (Fp1, Fp2, AF3, AF4, F3 and F4), whereas Cao2018 comprised 58 sessions and 4 single-electrode selections (Fp1, Fp2, F3 and F4). Because each selection contained exactly one channel, these values required no best-channel-per-session oracle, unlike every other condition in the article, and so corresponded directly to what a deployed single-electrode system would produce.

Consistent with the channel-subset results, the complete Proposed-Med pipeline retained most of the full-montage performance when operated on a single frontopolar electrode, whereas performance dropped sharply for single electrodes outside the frontopolar pair. The best single-electrode F_1 reached 83.73% for Fp1 on Internal (vs. 86.32% for the same electrode inside the full montage) and 77.65% for Fp1 on Cao2018 (vs. 79.68%), a gap of only 2–3 percentage points. Outside the frontopolar pair, the gap to the full montage widened: on Internal, single-electrode F_1 was 75.16% for AF4 (vs. 80.32%), 67.59% for AF3 (vs. 73.76%), 39.85% for F3 (vs. 45.65%), and 39.98% for F4 (vs. 45.23%), while on Cao2018 it was 49.30% for F3 (vs. 59.51%) and 49.03% for F4 (vs. 57.99%) (Figure 5).

The size of this gap tracked distance from the frontal pole rather than being uniform across electrodes: on Internal, the shortfall from the full montage widened from 2.59 percentage points at Fp1 and 3.45 at Fp2 to 5.16 at AF4, 6.17 at AF3, 5.79 at F3 and 5.26 at F4; on Cao2018, it widened from 2.03 points at Fp1 to 10.21 at F3 and 8.96 at F4. Because these single-electrode values carry no channel-selection oracle, they indicate that a deployed single-electrode system built on Fp1 or Fp2 would lose only a few points of F_1 relative to the full montage, whereas one built on the weaker frontal-adjacent electrodes would give up a much larger share of the achievable performance.

TABLE 2. Single-channel detection performance collapsed to coarse scalp regions (Proposed-Med, median centre). Each region row averages the per-channel macro F_1 over the electrodes it contains. The frontopolar, midline/outside, and unassigned groups are omitted; only the frontal, central, parietal, and occipital regions are shown. n is the number of electrodes in the region. F_1 is reported as a percentage.

Dataset	Region	n	F_1 (%)
Internal	frontal	4	61.24
	central	6	14.80
	parietal	4	1.94
	occipital	8	2.30
Cao2018	frontal	2	58.75
	central	6	38.13
	parietal	2	18.70
	occipital	6	8.24

B. EXPERIMENT 2: STABILITY ACROSS EPOCH DURATIONS

Epoch duration is routinely chosen on paradigm-specific or theoretical grounds and is rarely optimised for the blink detector itself. In driving studies, epoch lengths ranging from

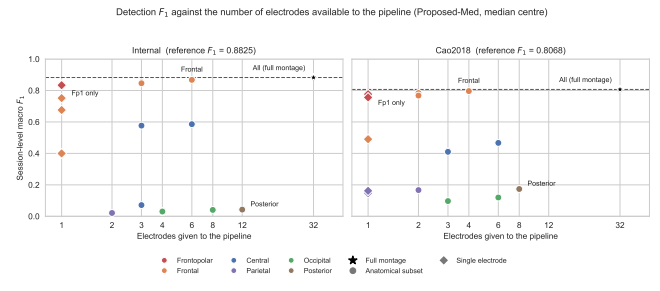


FIGURE 6. Detection F_1 against the number of electrodes given to the pipeline. Each point is one channel subset; the dashed line marks the full-montage reference.

TABLE 3. Best-channel selection frequencies pooled over the four conditions (best-channel-per-session). The 5 most frequently selected electrodes are listed per dataset, with the count and the fraction of all session $\times$ condition selections.

Dataset	Summary	Frequencies
Internal	Channels	Fp2 75/184 (0.41); Fp1 67/184 (0.36); AF4 14/184 (0.08); AF3 11/184 (0.06); Fz 4/184 (0.02)
Cao2018	Channels	Fp1 119/232 (0.51); Fp2 57/232 (0.25); F8 11/232 (0.05); Fz 11/232 (0.05); F7 9/232 (0.04)

a few seconds to several minutes appear in the literature. A robust pipeline should produce stable event-level F_1 across this range, because the underlying physiology of blinks does not change with the epoch grid imposed during analysis.

The pipeline was evaluated under epoch durations of 10, 20, 30 (reference), 40, 50, 60 and 120 seconds. For each duration the full pipeline is re-run from scratch, since epoch-level peak-to-peak statistics change with epoch length. All values are computed on the full 32-channel montage at the best-channel-per-session operating point, which is the same channel set and operating point reported in Experiments 1 and 3; the reduced frontal sub-montages that the sweep also covers are excluded here so that epoch duration is the only quantity that varies across rows. The 30 s row therefore reproduces the corresponding Proposed-Med values of those experiments exactly. Table 4 reports macro-averaged F_1 and two-tailed Wilcoxon p -values relative to the 30-second reference, Bonferroni-corrected over the six non-reference durations.

Across the seven epoch durations, the proposed algorithm maintained consistent pooled performance and similarly stable performance on Cao2018, although its Internal results declined significantly relative to the 30 s reference at the three longest durations (Table 4, Figure 7). Pooled macro F_1 varied by only 0.39 percentage points, from 83.67% at 120 s to 84.06% at 20 s. None of the six alternative durations differed significantly from the 30 s reference after Bonferroni correction, and the smallest corrected p -value was observed at 120 s ($p = 0.067$). Performance on Cao2018 was also stable, with an F_1 spread of 0.46 percentage points and no significant difference at any tested duration. Although 120 s produced the highest F_1 of 81.12%, exceeding the reference by 0.44 percentage points, this value did not differ significantly from

it ($p = 0.659$). Internal showed a less consistent pattern. The 30 s reference yielded the highest Internal F_1 , 88.25%, and relative to it F_1 fell to 87.93% at 50 s ($p < 0.001$), 87.73% at 60 s ($p = 0.002$), and 86.88% at 120 s ($p < 0.001$). All three differences remained significant after Bonferroni correction, whereas 10, 20 and 40 s did not differ from the reference. Significance tracked the consistency of the change rather than its magnitude. The 0.32-percentage-point mean drop at 50 s was significant because it recurred across sessions, 35 of 46 falling below the reference with a median change of -0.40 percentage points, whereas the larger 0.54-percentage-point mean drop at 10 s was not ($p = 1.000$): there sessions split 22 below against 24 above the reference and the median session improved by 0.18 percentage points, so the mean was carried by a minority of sessions with large declines. The evidence therefore supports stable pooled and Cao2018 performance over the full 10–120 s range and stable Internal performance from 10 to 40 s, but it does not support equivalence on Internal at 50 s and beyond.

TABLE 4. Best-channel-per-session macro- F_1 of Proposed-Med across epoch durations, computed on the full 32-channel montage: for every session the single electrode with the highest F_1 within that montage run is selected, then averaged over sessions. This is the same operating point and channel set reported in Experiments 1 and 3, so the 30 s row reproduces those values exactly. p -values (two-tailed Wilcoxon on session-level F_1 , Bonferroni-corrected over 6 non-reference durations) compare each duration against the 30 s reference. Macro F_1 is reported as a percentage. Bold marks the best duration within each block.

Dataset	Epoch duration	n	Macro F_1 (%)	p vs. 30 s
Internal	10 s	46	87.71	1.000
	20 s	46	88.21	1.000
	30 s	46	88.25	reference
	40 s	46	88.14	0.610
	50 s	46	87.93	< 0.001
	60 s	46	87.73	0.002
	120 s	46	86.88	< 0.001
Cao2018	10 s	58	80.77	1.000
	20 s	58	80.76	1.000
	30 s	58	80.68	reference
	40 s	58	80.66	1.000
	50 s	58	80.74	1.000
	60 s	58	81.05	0.402
	120 s	58	81.12	0.659
Pooled	10 s	104	83.84	1.000
	20 s	104	84.06	1.000
	30 s	104	84.03	reference
	40 s	104	83.97	0.933
	50 s	104	83.92	0.523
	60 s	104	84.01	1.000
	120 s	104	83.67	0.067

C. EXPERIMENT 3: EFFECT OF THE THRESHOLD ESTIMATOR AT STAGE B

Stage B estimates the sample-level threshold from samples pooled across the suspicious epoch set $\mathcal{S}$. Because $\mathcal{S}$ is selected precisely because it contains high-amplitude excursions, the pooled distribution is not symmetric around a clean baseline such that it is contaminated by the very peaks the threshold is meant to capture. The mean and standard devi-

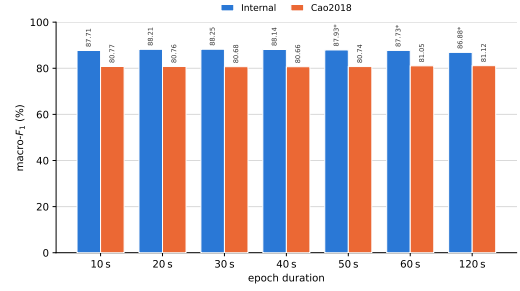


FIGURE 7. Macro-averaged F_1 of Proposed-Med across epoch durations, on the full 32-channel montage at the best-channel-per-session operating point. Bars are drawn on the full 0–100% axis, so each bar is labelled with its value; an asterisk marks the durations that differ significantly from the 30 s reference after Bonferroni correction (Table 4).

ation are sensitive to these extreme values whereas the median and MAD are not. The hypothesis is that the proposed-Med will outperform Proposed-Mean on precision—because the MAD-based threshold is less inflated by within-epoch peaks—while achieving comparable or superior recall.

Within the two proposed configurations, the Stage-B threshold estimators produced similar overall performance, although the direction of the metric changes did not support the stated precision hypothesis (Table 5). On Internal, Proposed-Med achieved precision, recall, and F_1 of 88.03%, 90.14%, and 88.25%, compared with 88.40%, 88.76%, and 87.42% for Proposed-Mean. The corresponding changes were -0.37 percentage points in precision, $+1.38$ in recall, and $+0.83$ in F_1 . On Cao2018, Proposed-Med likewise reduced precision by -1.16 percentage points while increasing recall by $+1.89$ and F_1 by $+0.38$. Pooled performance followed the same pattern, with precision decreasing from 82.09% to 81.27%, recall increasing from 87.89% to 89.56%, and F_1 increasing from 83.45% to 84.03%. However, the two estimators were statistically indistinguishable under the two-sided Bonferroni-corrected Wilcoxon test ($p = 1$). Proposed-Med exceeded Proposed-Mean in 46/104 sessions, was equal in 4, and was lower in 54, indicating a per-session outcome close to even.

TABLE 5. Effect of the Stage-B threshold estimator on the Internal and Cao2018 driving-EEG corpora at 30 s epochs. Proposed-Med (median/MAD) and Proposed-Mean (mean/SD) are summarised at the best-channel-per-session operating point (for every session the single channel or frontal sub-montage with the highest F_1 is selected, then averaged over sessions). Macro-averaged F_1 is reported as a percentage, per dataset and pooled over all 104 sessions. Best F_1 per block in bold. By a paired Wilcoxon signed-rank test on session-level F_1 (Bonferroni-corrected over six condition pairs), Proposed-Med and Proposed-Mean are statistically indistinguishable ($p = 1.00$).

Dataset	Condition	F_1 (%)
Internal	Proposed-Mean	87.42
	Proposed-Med	88.25
Cao2018	Proposed-Mean	80.30
	Proposed-Med	80.68
Pooled	Proposed-Mean	83.45
	Proposed-Med	84.03

D. EXPERIMENT 4: STRATEGY COMPARISON

In driving and other epoch-structured paradigms, it is common to apply a continuous-signal blink detector to the raw or minimally preprocessed recording and subsequently map the detected blinks back onto the epoch grid. Both BLINKER and the standard MNE pipeline follow this approach, in which channel-specific thresholds are derived from a signal pool containing both blink-heavy and blink-free epochs. The central premise of the proposed pipeline is that pooling these heterogeneous epochs dilutes the amplitude statistics used for threshold estimation, thereby distorting threshold placement and shifting the resulting precision–recall trade-off.

Accordingly, we hypothesise that naive concatenation will produce systematically lower session-level F_1 scores than the epoch-aware pipeline because of a less favourable precision–recall trade-off. To evaluate this hypothesis, all four pipelines (i.e., BLINKER, MNE, Proposed-Med, and Proposed-Mean) are applied to every session using 30-second epochs and all available channels. All 32 channels are included to ensure a fair comparison with BLINKER, which, by default, processes all channels provided to it and requires the user to select a representative blink channel. For consistency across methods, all pipelines are therefore supplied with the same 32 channels, and the channel yielding the highest F_1 score is selected for subsequent analysis. The per-session macro- F_1 , precision, and recall are summarised in Table 6.

In the strategy comparison, both proposed configurations achieved higher macro- F_1 scores than BLINKER and MNE on Internal, Cao2018, and the pooled evaluation (Table 6). Proposed-Med achieved macro- F_1 scores of 88.25%, 80.68%, and 84.03%, respectively, whereas Proposed-Mean achieved 87.42%, 80.30%, and 83.45%. The corresponding scores were 76.54%, 69.40%, and 72.56% for BLINKER and 66.64%, 56.27%, and 60.86% for MNE.

This performance gap is reflected in the precision–recall profiles shown in Figure 8. BLINKER achieved the highest recall among all conditions on Internal, Cao2018, and the pooled evaluation, with values of 95.18%, 97.15%, and 96.27%, respectively, but also exhibited the lowest precision, at 67.94%, 56.96%, and 61.82%. Wilcoxon signed-rank tests on session-level F_1 , with Bonferroni correction across the six pairwise comparisons, showed significant differences between each proposed configuration and each baseline. Proposed-Med versus BLINKER yielded $p = 1.32 \times 10^{-11}$, and Proposed-Med versus MNE yielded $p = 6.51 \times 10^{-9}$. Similarly, Proposed-Mean versus BLINKER yielded $p = 4.97 \times 10^{-10}$, while Proposed-Mean versus MNE yielded $p = 9.94 \times 10^{-9}$. In contrast, neither within-pair comparison was significant: BLINKER versus MNE yielded $p = 0.812$, whereas Proposed-Mean versus Proposed-Med yielded $p = 1$. Furthermore, no channel group in either dataset produced an inversion in which a baseline matched or exceeded the performance of Proposed-Med (Table 7).

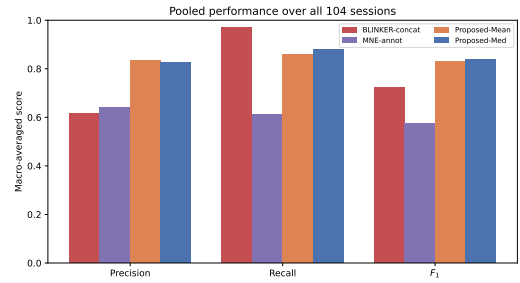


FIGURE 8. Macro-averaged precision, recall, and F_1 for the four conditions over all sessions.

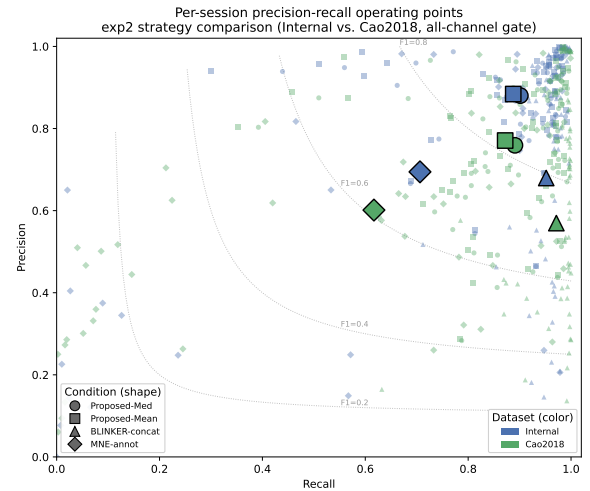


FIGURE 9. Per-session precision–recall operating points, Internal and Cao2018 shown separately.

E. OPERATING POINTS

Across the pooled and corpus-specific analyses, the precision–recall operating points showed a consistent ordering among the proposed and baseline conditions (Figure 9 and Figure 10). In the pooled analysis, Proposed-Med and Proposed-Mean occupied the high-precision, high-recall region, with precision and recall of 81.27% and 89.56% for Proposed-Med and 82.09% and 87.89% for Proposed-Mean, respectively. BLINKER-concat occupied the high-recall, low-precision region, with recall of 96.27% and precision of 61.82%, whereas MNE-annot was lower on both measures, with precision of 64.25% and recall of 65.64%. The same ordering was observed within each corpus. On Internal, the proposed conditions had precision of 88.03% and 88.40% and recall of 88.76% and 90.14%, whereas BLINKER-concat combined 95.18% recall with 67.94% precision and MNE-annot achieved 69.44% precision and 70.63% recall. On Cao2018, the corresponding pattern remained, with the proposed conditions retaining higher precision and recall than MNE-annot, while BLINKER-concat retained the highest recall and lower precision.

TABLE 6. Strategy comparison on the Internal and Cao2018 driving-EEG corpora at 30 s epochs. Each condition is summarised at its best-channel-per-session operating point (for every session the single channel or frontal sub-montage with the highest F_1 is selected, then averaged over sessions); the same rule is applied to all four conditions. Macro-averaged precision, recall and F_1 are reported as percentages, per dataset and pooled over all 104 sessions. Best F_1 per block in bold. By paired Wilcoxon signed-rank tests on session-level F_1 (Bonferroni-corrected over six pairs), Proposed-Med significantly exceeds BLINKER-concat ($p < 10^{-9}$) and MNE-annot ($p < 0.001$) but not Proposed-Mean ($p = 1.00$).

Dataset	Condition	Precision (%)	Recall (%)	F_1 (%)
Internal	BLINKER-concat	67.94	95.18	76.54
	MNE-annot	69.44	70.63	66.64
	Proposed-Mean	88.40	88.76	87.42
	Proposed-Med	88.03	90.14	88.25
Cao2018	BLINKER-concat	56.96	97.15	69.40
	MNE-annot	60.13	61.68	56.27
	Proposed-Mean	77.07	87.20	80.30
	Proposed-Med	75.91	89.09	80.68
Pooled	BLINKER-concat	61.82	96.27	72.56
	MNE-annot	64.25	65.64	60.86
	Proposed-Mean	82.09	87.89	83.45
	Proposed-Med	81.27	89.56	84.03

TABLE 7. Channel groups and datasets where a baseline algorithm equals or exceeds Proposed-Med in best-channel-per-session macro- F_1 . Values are percentages; F_1 is in percentage points.

Dataset	Selection	Baseline	BL- F_1 (%)	Prop- F_1 (%)	ΔF_1 (pp)
No inversions — Proposed-Med leads on every channel group.					

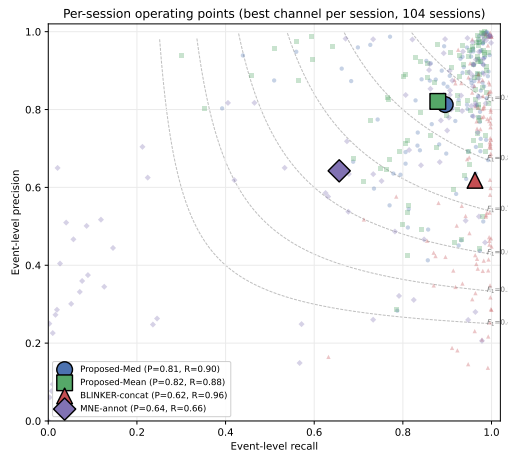


FIGURE 10. Per-session operating points pooled over both corpora, with condition means and iso- F_1 contours.

IV. DISCUSSION

Experiment 1 showed that useful detection information was concentrated at the frontopolar sites and that reduced frontal configurations retained most of the full-montage performance. The frontal subset achieved macro- F_1 values of 86.76% on Internal and 79.61% on Cao2018, compared with 88.25% and 80.68% for the respective full-montage references; neither difference was statistically significant after correction ($p = 1.000$ in both corpora). Within the full-montage run, the best individual electrode was Fp2 on Internal ($F_1 = 86.82\%$) and Fp1 on Cao2018 ($F_1 = 79.68\%$). When the complete pipeline was instead run on one electrode

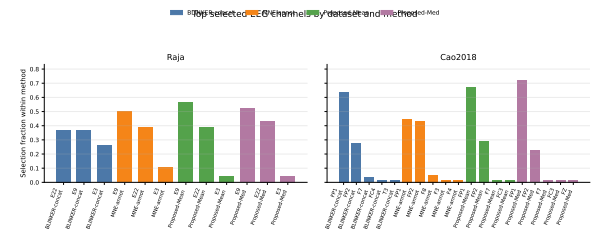


FIGURE 11. Best-channel selection frequency, pooled over the four conditions.

TABLE 8. Stability of the best-channel choice across the four conditions. Full agreement is the fraction of sessions where all four conditions select the same best channel; per-condition agreement is the mean fraction of the other three conditions selecting the same best channel.

Dataset	Sessions	Full agreement	Mean pairwise	BLINKER-concat	MNE-annot	Proposed-Med
Internal	46	9/46 (0.20)	0.49	0.44	0.33	0.50
Cao2018	58	6/58 (0.10)	0.41	0.40	0.22	0.50
Pooled	104	15/104 (0.14)	0.44	0.42	0.27	0.50

alone, the best single-electrode result was obtained at Fp1 in both corpora, reaching 83.73% on Internal and 77.65% on Cao2018. Performance declined sharply for non-frontal subsets, confirming that the frontal subsets were sufficient under the tested design, although the subset analysis did not establish that every frontal channel was necessary. This concentration is compatible with the strong projection of blink activity to the prefrontal scalp and with earlier single-channel and forehead-EEG studies [10, 12, 15, 16, 21, 23, 28–30]. The findings therefore motivate further evaluation of reduced frontopolar systems for wearable or in-vehicle use, without establishing that such systems will reproduce the retrospective full-montage operating point in deployment.

The channel-choice analyses reported with Experiment 1 further showed that this spatial concentration did not translate into a stable choice of one electrode. Across the four conditions, Fp1 and Fp2 accounted for most selections: Internal selected Fp2 in 75/184 cases and Fp1 in 67/184, whereas

Cao2018 selected Fp1 in 119/232 and Fp2 in 57/232. Nevertheless, all four conditions selected the same electrode in only 9/46 Internal sessions and 6/58 Cao2018 sessions, giving pooled full agreement in 15/104 sessions. The results therefore support stability at the level of the frontopolar region, but not interchangeability of Fp1 and Fp2 at the session level. Differences in ocular projection, electrode placement, and signal amplitude offer plausible explanations for this variability [24, 31], but these factors were not isolated in the present analysis. Moreover, because the reported best channel was selected retrospectively using reference F_1 , the results do not demonstrate a deployable adaptive channel-selection rule; they instead motivate evaluating a label-free rule for choosing between frontopolar channels.

Experiment 2 indicated that Proposed-Med was insensitive to epoch duration in the pooled analysis, where macro- F_1 spanned only 0.56 percentage points across the seven tested durations and none differed significantly from the 30-second reference after correction (Table 4; Figure 7). Cao2018 was similarly stable, with an F_1 spread of 0.71 percentage points and no significant difference at any duration. This pattern is useful because epoch duration is commonly determined by the experimental paradigm rather than by the blink detector. The corpus-specific results nevertheless qualify the pooled finding: on Internal, F_1 declined from 88.82% at 30 seconds to 88.60% at 50 seconds ($p < 0.001$), 88.37% at 60 seconds ($p = 0.002$), and 87.20% at 120 seconds ($p < 0.001$), and all three differences remained significant after correction. Thus, the evidence supports stable pooled performance and no detected change from 10 to 40 seconds in either corpus, but it does not establish equivalence across the complete 10–120-second range within every dataset. Longer epochs may alter the composition of the signal used during screening and threshold estimation, but the present analysis did not test that mechanism directly.

Experiment 3 showed that the choice between the median/MAD and mean-based Stage-B estimators did not materially explain the performance of the proposed pipeline. The median/MAD centre was intended to reduce the influence of extreme values remaining in the suspicious-epoch pool, consistent with the robustness of median-based estimators to outlying observations [32, 33]. However, pooled precision changed from 82.09% with Proposed-Mean to 81.27% with Proposed-Med, while recall changed from 87.89% to 89.56% and F_1 from 83.45% to 84.03%. The estimators were statistically indistinguishable after correction ($p = 1$): Proposed-Med exceeded Proposed-Mean in 46/104 sessions, was equal in 4, and was lower in 54. The median centre therefore did not provide the hypothesised precision advantage and cannot be credited as a demonstrated source of improvement. It remains a defensible default on robustness grounds, but the measured comparison supports similar overall performance from the two estimators.

Experiment 4 showed that the epoch-aware proposed pipeline outperformed both comparison methods across the two driving corpora. Proposed-Med achieved a pooled macro-

F_1 of 84.03%, compared with 72.56% for BLINKER-concat and 60.86% for MNE-annot (Table 6). BLINKER-concat represented an established blink-specific method applied to the concatenated recording [8], whereas MNE-annot represented the default amplitude-annotation routine from MNE-Python.

The corresponding session-level differences were statistically significant for both BLINKER-concat ($p = 1.32 \times 10^{-11}$) and MNE-annot ($p = 6.51 \times 10^{-9}$), and neither baseline equalled or exceeded Proposed-Med in any channel group on either dataset. These comparisons establish the advantage of the complete proposed configuration under the common evaluation protocol. Because the experiment compared complete detection strategies rather than a screening-on versus screening-off ablation of an otherwise identical detector, it does not by itself attribute the improvement exclusively to epoch screening.

The precision–recall profile is consistent with, but does not prove, the proposed explanation that heterogeneous epoch composition affects a threshold derived from the concatenated recording. BLINKER-concat had the highest pooled recall of 96.27% but substantially lower precision of 61.82%, whereas Proposed-Med combined recall of 89.56% with precision of 81.27%. BLINKER-concat therefore operated at a more permissive point, producing greater sensitivity together with more false detections. Variation in ocular-artifact amplitude across subjects, electrodes, and recording sessions makes sensitivity to the signal pool plausible [6, 8, 11, 24]. However, the reported experiments did not directly measure epoch composition, the resulting threshold shift, or the isolated effect of enabling and disabling epoch screening. The supported interpretation is consequently that the proposed pipeline achieved a more balanced operating point; a causal attribution to reduced threshold dilution requires a dedicated ablation.

These findings are subject to several limitations. The benchmark was restricted to two driving corpora, so generalisation to other experimental paradigms or clinical populations was not tested. Event detection was reported under one overlap-based matching rule, and sensitivity to alternative event-matching criteria was not evaluated; absolute F_1 values should therefore not be compared across studies without accounting for their scoring rules [34]. Performance was not reported separately by blink duration, so detection of prolonged eye closures, which is particularly relevant to drowsiness assessment, remains unestablished. The evaluation also depends on the available reference annotations and their operational definitions of blink onset and offset, boundaries that can vary with signal representation and annotation criteria [1]. In addition, the headline strategy comparison used a retrospective best-channel-per-session rule based on reference F_1 ; this common oracle was applied fairly to every condition, but its scores do not represent prospective channel selection without labels. Finally, the proposed thresholding method was not benchmarked against learned detectors such as transformer-, vision-, or other deep-learning approaches [35–37]. Its demonstrated strengths are therefore limited to the methods and evaluation protocol examined here.

Future work should therefore evaluate prolonged eye closures separately, test prospective channel selection, and broaden the benchmark. Reporting normal blinks and prolonged closures independently would establish whether performance is retained for the sustained closures most relevant to drowsiness assessment [3, 5]. A prospective study should also replace the reference-based best-channel oracle with a fixed frontopolar electrode or a label-free quality criterion and quantify the resulting performance loss. Evaluation beyond driving data, including paradigms organised into task blocks, trials, or other natural segments, would test whether the observed robustness extends to different blink morphologies, cognitive demands, and recording conditions. Finally, the epoch-screening idea could be assessed as a front end to learned blink or drowsiness detectors [38–40]; such a hybrid remains a proposal and was not evaluated in the present study.

V. CONCLUSION

Taken together, the findings indicate that screening epoch-structured EEG before estimating sample-level thresholds provides more reliable event-level blink-region detection than deriving a single threshold from a concatenated recording under the evaluated driving conditions. The proposed median-based configuration outperformed the examined baselines across both driving corpora, supporting epoch screening as the principal methodological contribution of the pipeline. The spatial analysis further showed that frontopolar electrodes retained most of the full-montage performance, while the epoch-duration analysis indicated broadly stable pooled performance despite some corpus-specific sensitivity to longer epochs. Collectively, these findings support an interpretable and spatially sparse approach to epoch-aware blink detection that can accommodate common segmentation choices without requiring a dense electrode montage. The evaluation deliberately focused on driving data to establish the method's performance within its intended fatigue-driving context. Consequently, its generalizability beyond this domain and its suitability for practical deployment remain to be established. Future work should validate the pipeline across independent driving datasets and recording systems before examining its transfer to other epoch-structured paradigms. Overall, this study positions epoch screening as a practical foundation for improving blink-detection threshold calibration in heterogeneous, epoch-structured EEG.

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